Brain Tumor Classification using Deep Learning with MRI Images

Project Overview

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Current Challenges

The Need for Deep Learning in Brain Tumor Classification

- Time-Intensive Process: Radiologists spend considerable time analyzing complex scans, leading to diagnostic delays.
- Subjectivity and Variability: Manual analysis is subjective and can vary between experts, causing inconsistencies.
- Increasing Workload: The growing volume of medical imaging data burdens healthcare professionals.

Deep learning offers a solution to enhance objectivity, efficiency, and support clinical decisions.

BRISC 2025 Dataset Description

Source

BRISC 2025 Kaggle dataset

Content

Brain MRI images

Classes

Includes Glioma, Meningioma, Pituitary tumors, and No tumor cases.

Challenge

Multi-class classification with moderate class imbalance and real-world variability.

Data Split

Standard train/validation/test strategy applied.

The dataset presents a realistic multi-class classification challenge within neuroimaging.

Methodology: Model Architecture

Core Model

Utilizes EfficientNetB3, a pre-trained state-of-the-art convolutional neural network.

Training Strategy

Employs a full fine-tuning approach for the model.

Input Resolution

Processes input data at a resolution of 300x300 pixels.

Custom Classification Head

Features a tailored classification head including BatchNorm, Dense, and Dropout layers for specific task requirements.

Training Strategy

Loss Function

Focal Loss is used to address class imbalance issues.

Optimizer

The Adam optimizer is employed with a learning rate of 1e-4.

Class Weighting

Class weighting is applied to emphasize the glioma class.

Regularization

Early stopping and learning rate scheduling are utilized to optimize performance and prevent overfitting.

Methodology: Data Augmentation

Enhancing Model Generalization and Robustness

Geometric Transformations

Applies transformations such as random flipping, rotation, and zooming to the data.

Color Space Adjustments

Modifies color properties like contrast and brightness to augment the dataset.

Overall Performance Results

Validation Accuracy

Achieved approximately 96.7% accuracy on the validation dataset.

Test Accuracy

Achieved approximately 62.9% accuracy on the unseen test dataset.

Generalization Gap

A significant difference between validation and test accuracy indicates a generalization gap.

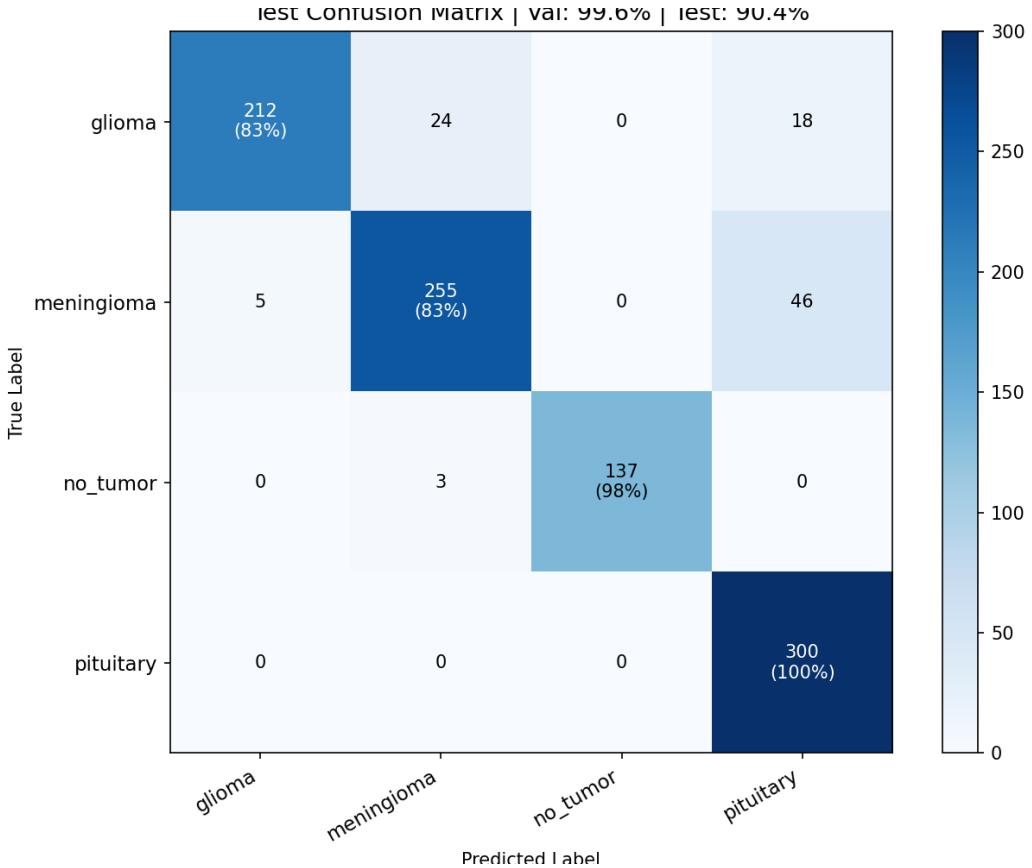
Potential Overfitting

The observed gap suggests the model might be overfitting to the validation data.

The significant generalization gap implies potential overfitting, requiring further investigation and model refinement to improve performance on unseen data.

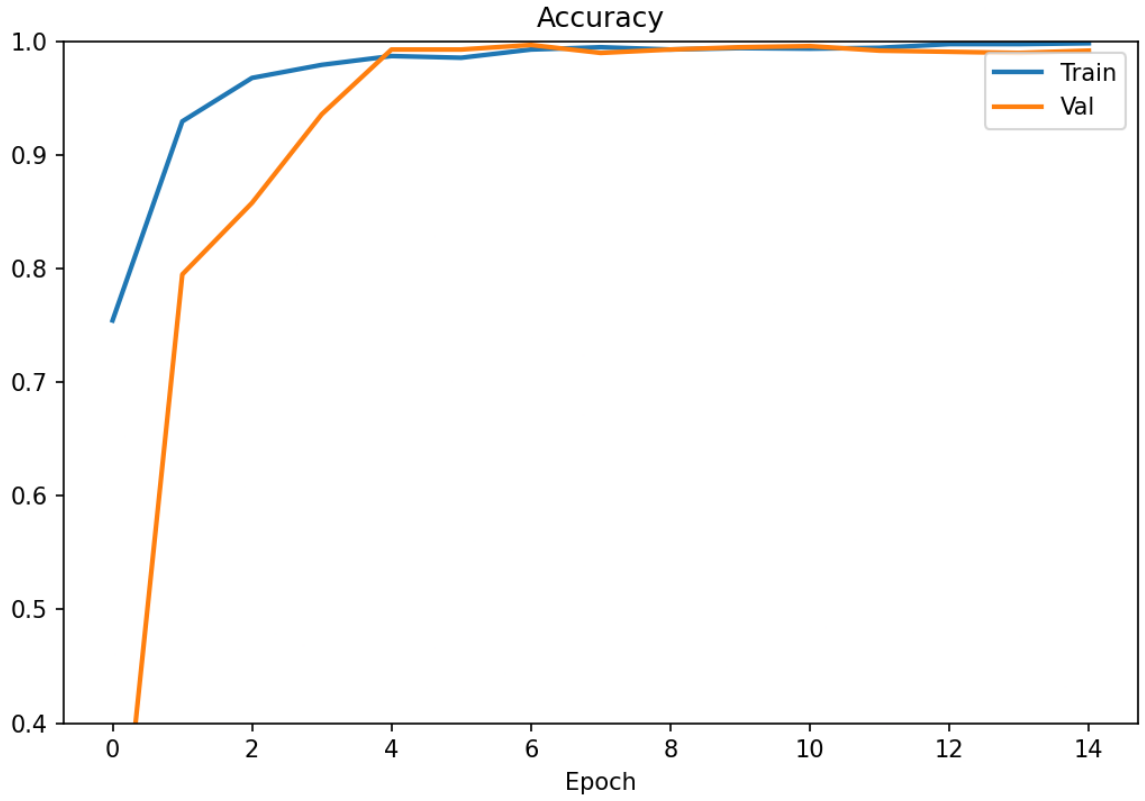
Confusion Matrix Analysis: Tumor Classification Performance

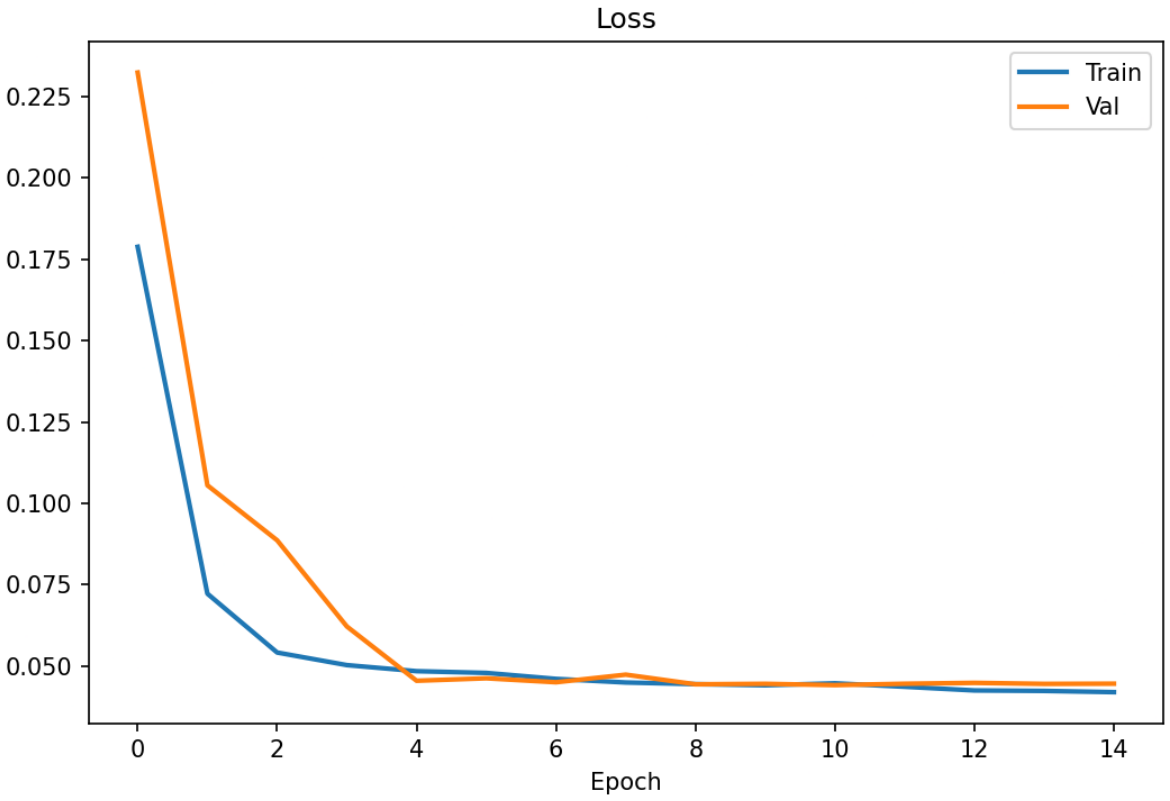
Highlighting accuracy variations and common misclassifications.



Model Performance: Training and Validation Curves

Analysis of Accuracy and Loss Trends





Strong model fitting with good generalization capabilities.

Grad-CAM Visualization Results

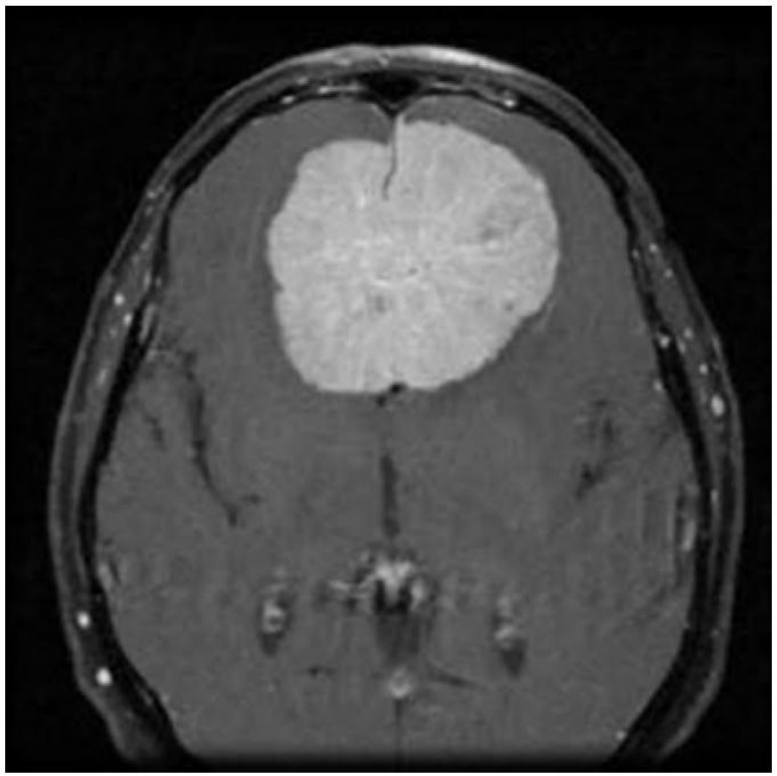
Understanding Model Attention

Correct Focus

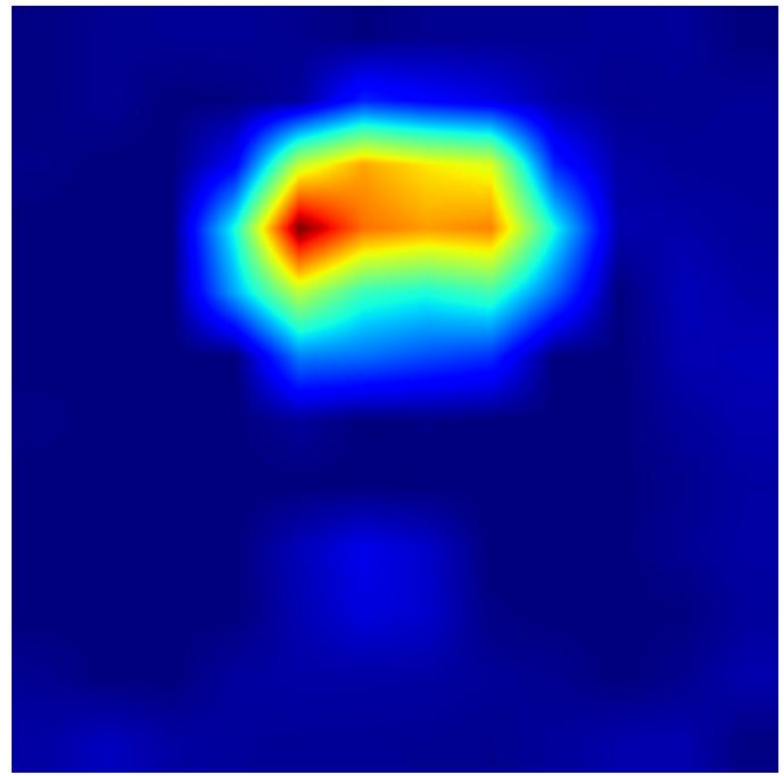
The model accurately focuses on tumor regions in several instances, demonstrating a good understanding of relevant features.

Grad-CAM visualizations provide insight into the model's decision-making process, highlighting areas of attention and helping to explain observed classification patterns.

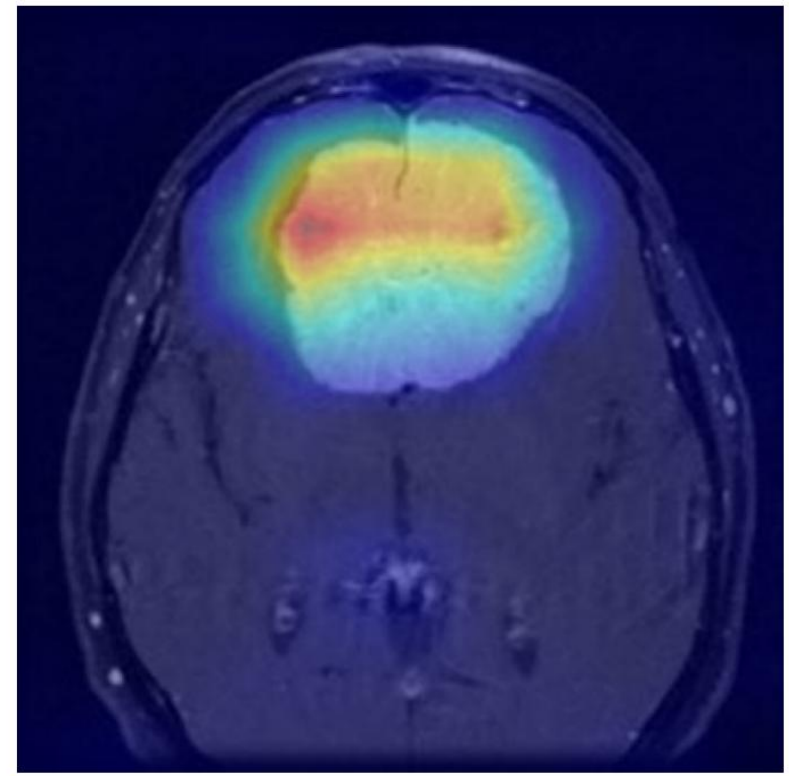
Original



Grad-CAM



Vorhersage: meningioma
Konfidenz: 98.7%



Class-wise Performance Analysis

Model Performance Across Tumor Categories

Class	Precision	Recall	F1-score
Glioma	0.93	0.90	0.91
Meningioma	0.94	0.84	0.89
No Tumor	0.96	0.99	0.98
Pituitary	0.87	1.00	0.93

The model exhibits consistent performance, with minor areas for improvement in distinguishing between potentially similar tumor types.

ROC Curve Analysis (Multiclass)

Model Discrimination Capability Across Brain Tumor Classes

AUC Scores per Class

Area Under the Curve (AUC) scores for Glioma (0.99), Meningioma (0.98), No Tumor (1.00), and Pituitary (1.00) tumor classes.

ROC Curve Characteristics

All ROC curves show a rapid increase towards the top-left, indicating high sensitivity and specificity.

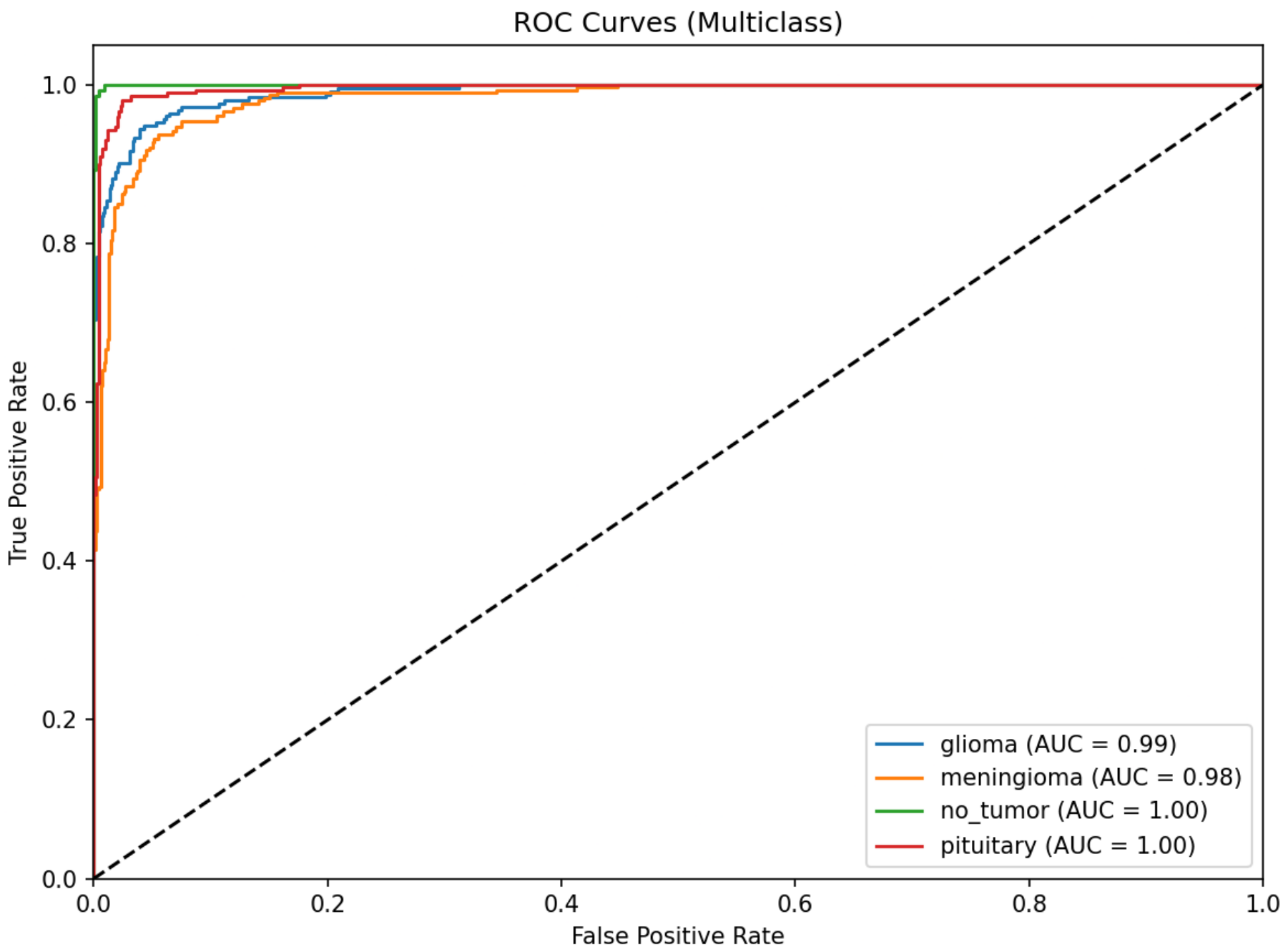
Class Separability

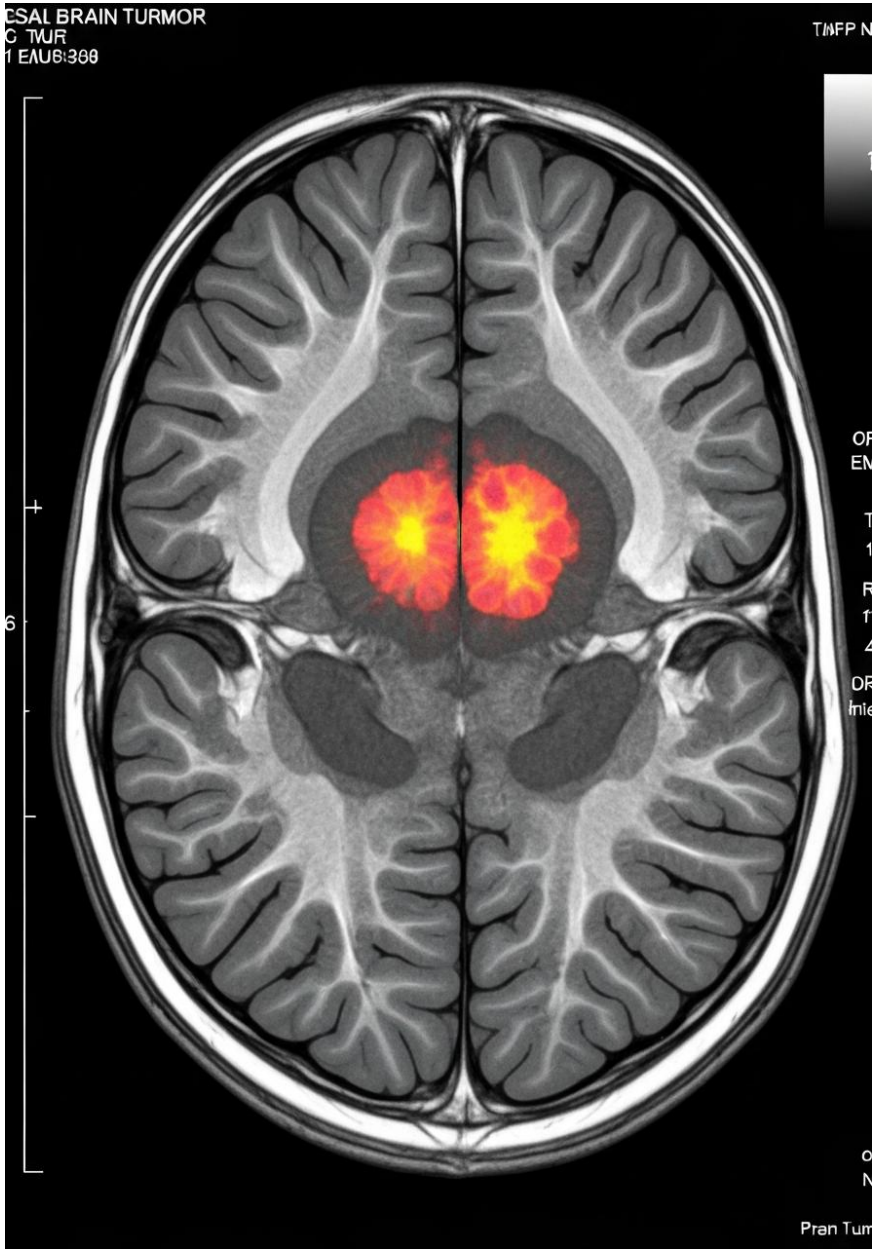
Near-perfect AUC scores suggest excellent separation between classes.

Performance Across Thresholds

The model demonstrates strong performance irrespective of classification thresholds.

The model exhibits outstanding discrimination capability, effectively distinguishing between all tumor classes.


ROC Curves (Multiclass)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
glioma (AUC = 0.99)
meningioma (AUC = 0.98)
no_tumor (AUC = 1.00)
pituitary (AUC = 1.00)



Performance Discrepancy

Interpretation: Key Insight

The significant discrepancy between validation and test performance highlights limited generalization capability.

Potential causes include a distribution shift between datasets or the model relying on non-robust features that do not generalize well to unseen data.

Limited generalization capability due to performance discrepancy.

Interpretation: Strengths

Strong Performance on Specific Tumor Types

Demonstrated effectiveness on particular conditions, such as pituitary tumors.

Effective Feature Extraction

Leverages the power of EfficientNet for robust extraction of relevant features.

Explainability through Grad-CAM

Provides insights into the model's decision-making process using Grad-CAM visualizations.

Interpretation: Limitations

Generalization Gap

A notable difference exists between the performance of the model on validation and test datasets.

Complex Tumor Types

The model exhibits reduced efficacy when applied to more intricate forms of tumors.

Dataset Sensitivity

The model's performance is significantly influenced by the specific characteristics of the dataset it is trained or tested on.

These limitations underscore the necessity for continued model refinement and validation.

Future Work

Enhancing Performance and Clinical Applicability

Data Integrity

Implement patient-level data splitting to prevent data leakage.

Data Augmentation

Utilize larger, multi-institutional datasets for improved generalizability.

Advanced Architectures

Explore state-of-the-art models like ConvNeXt and EfficientNetV2.

Model Adaptation

Investigate and implement domain adaptation techniques for broader applicability.

Ensemble Learning

Leverage ensemble methods to boost predictive accuracy and reliability.

Conclusion

Promise of Deep Learning

Deep learning shows significant potential for the automated classification of brain tumors using MRI scans.

Generalization Challenges

While current models perform well in validation, achieving reliable generalization in real-world clinical settings is a primary obstacle.

Future Research Needs

Continuous research and development are essential to ensure the clinical viability and ultimate reliability of these AI-driven diagnostic tools.

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